XUEWU CAI

Quantitative Research & Decision Support | Monte Carlo, Forecasting & Risk | MATLAB, Python, Java

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PROFESSIONAL SUMMARY

Financial engineering director with more than 20 years in quantitative research, financial modeling, forecasting and decision support. Applies Monte Carlo simulation, bootstrapping, statistical analysis and scenario-based risk measurement. Leads enterprise forecasting, capital planning and stress testing platform delivery, with experience translating analytical findings into Investment Committee reporting and transaction economics.

CORE COMPETENCIES

Quantitative Methods: Monte Carlo simulation, bootstrapping, volatility/correlation forecasts, scenario and stress analysis

Decision Support: portfolio risk, factor analysis, capital planning, CECL, DFAST, transaction economics, ROE

Programming & Computing: MATLAB, Python, Java, C/C++, SQL, AWS, IBM Symphony, QuantLib

Leadership & Reporting: technical leadership, multi-project delivery, financial model implementation, SEC/CFTC/OPERA reporting

PROFESSIONAL EXPERIENCE

Director, Financial Engineering — Freddie Mac

May 2014 – Present

McLean, VA | Credit Risk Transfer (STACR / ACIS)

- Own an enterprise forecasting platform for income statements, balance sheets and CECL estimates, supporting business planning, capital planning and DFAST stress testing.
- Direct development of modeling tools combining prepayment/default models, AWS and IBM Symphony computing, and Intex valuation analytics to assess bond performance and transaction economics.
- Lead engineering projects for STACR bond issuance and ACIS reinsurance within a CRT program protecting more than $1.6 trillion in mortgage loans.
- Deliver RPL and synthetic-structure deal memorandum, capital, economics and ROE analytics; use Copilot and harness engineering to support delivery.
- Manage the LIBOR-to-SOFR transition within CRT modeling tools and analytics.

Campbell & Company | Research Engineer and Engineering Roles

Mar 2003 – Apr 2014

Washington, DC–Baltimore Area

Research Engineer, Portfolio and Risk Team (2009–2014); Quantitative Finance Engineer (2006–2008); Senior Software Engineer (2003–2006).

- Developed volatility and correlation forecasting models in MATLAB using daily, intraday, implied-volatility and tick data.
- Applied statistical methods, Monte Carlo simulation and bootstrapping to investigate market stress, portfolio drawdowns, regime changes and correlation breakdowns using MATLAB, C/C++ and Java.
- Designed risk reporting on factor analysis, scenarios, stress tests, duration and margin for Investment Committee decisions; built an FX options pricing, simulation and risk engine using Visual C++, Boost and QuantLib.

Senior Software Engineer — Aquilent

Jun 1999 – Feb 2003

Java/Linux integration and data systems

- Built Java/Linux sensor integration systems with embedded Linux, JNI, serial communications and dynamic sensor discovery; developed Apache/Tomcat visualization portals and server-side satellite data components.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master’s Degree, Computer Science — University of Maryland Baltimore County

M.S., Computer Science & Engineering — Shanghai Jiao Tong University

Bachelor’s Degree, Applied Mathematics — Dalian University of Technology

Professional development: CAIA (Chartered Alternative Investment Analyst) Level I

LANGUAGES

Chinese: native or bilingual proficiency | English: full professional proficiency